Lab 10 Hashing Password

1. Continue from the previous project.
2. Using the same Web Form (Registration.aspx) as shown in Figure 1.

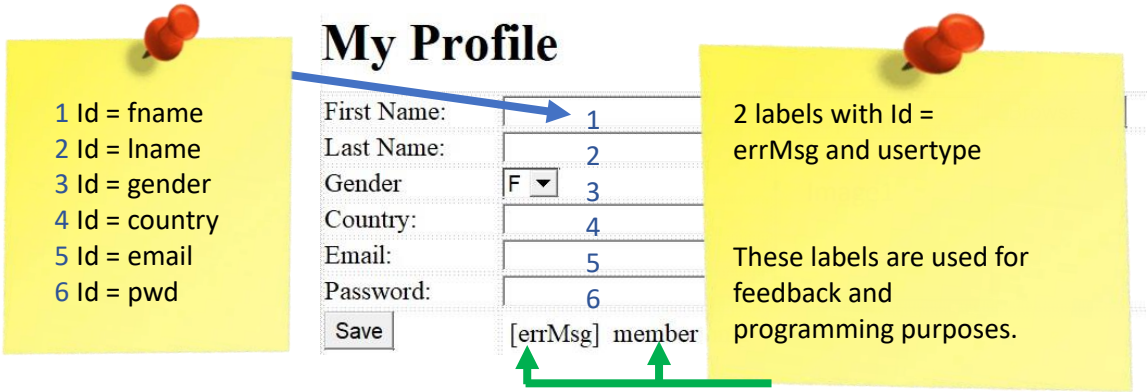
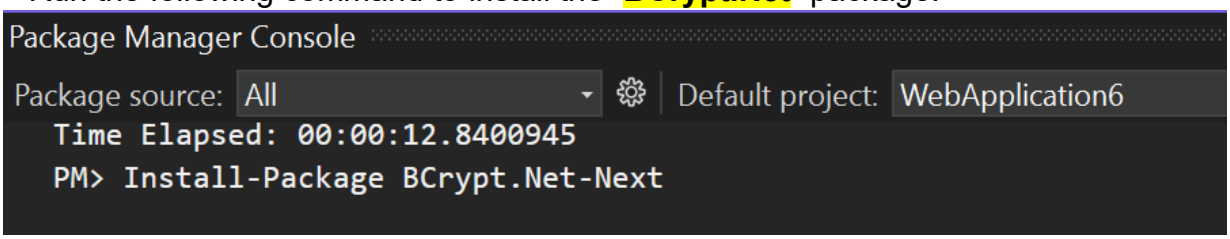


Figure 1: Manage Member Profile Page

Install Package

3. Goto > **Package Manager Console** (Tools > NuGet Package Manager > Package Manager Console).

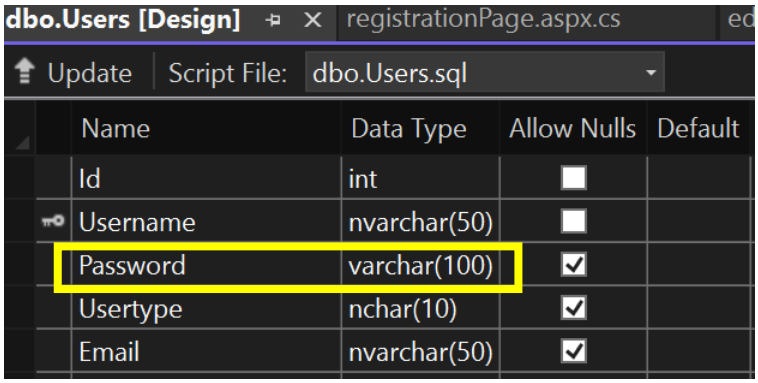
   Run the following command to install the `**BCrypt.Net**` package:



   Type in the following command after PM>

   Install-Package BCrypt.Net-Next

4. Open table definition:-



Change the size of length from 50 to 100.

Registration Page

5. Open the C# for registration (e.g. registration.cs). Put in the following code:

```
using BCrypt.Net;
```

```
//create record in a table called Users
string query1 = "insert into Users (Id, Username, Password, Usertype,
SqlCommand cmd1 = new SqlCommand(query1, con);
cmd1.Parameters.AddWithValue("@id", id.Text);
cmd1.Parameters.AddWithValue("@username",username.Text);

//hashing
var passwordManager = new PasswordManager();
string hashedPassword = passwordManager.HashPassword(pwd.Text);
Console.WriteLine($"Hashed Password: {hashedPassword}");

cmd1.Parameters.AddWithValue("@password", hashedPassword);
cmd1.Parameters.AddWithValue("@usertype", usertype.Text);
cmd1.Parameters.AddWithValue("@email", email.Text);
cmd1.ExecuteNonQuery();
```

```
public class PasswordManager
{
    // Method to hash the password
    1 reference
    public string HashPassword(string plainTextPassword)
    {
        string hashedPassword = BCrypt.Net.BCrypt.HashPassword(plainTextPassword);
        return hashedPassword;
    }
}
```

You may check your data entry:-

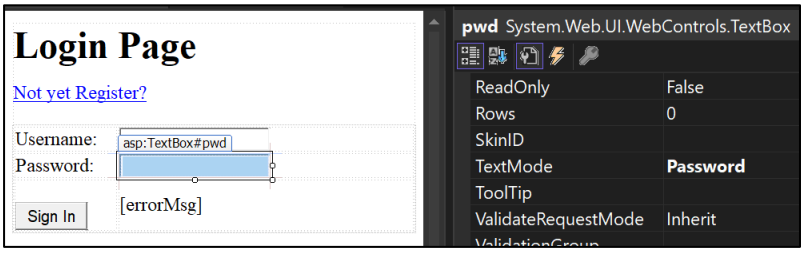
dbo.Users [Data] | login1.aspx | login1.aspx.cs | registrationPa

Max Rows: 1000

	Id	Username	Password	Usertype	Email
	4	ali	33445	admin	ali@abc.com
	2	aminah	1234	admin	aminah@ab...
	13	dragon	$2a$11$BO...	member	dragon@fru...
	5	flora	889900	member	flora@ddd.c...
	9	hilda	8899	admin	hilda@qq.co...

Login Page

1. Open the login.aspx file.



2. Insert the library as below:-

```
using BCrypt.Net;
```

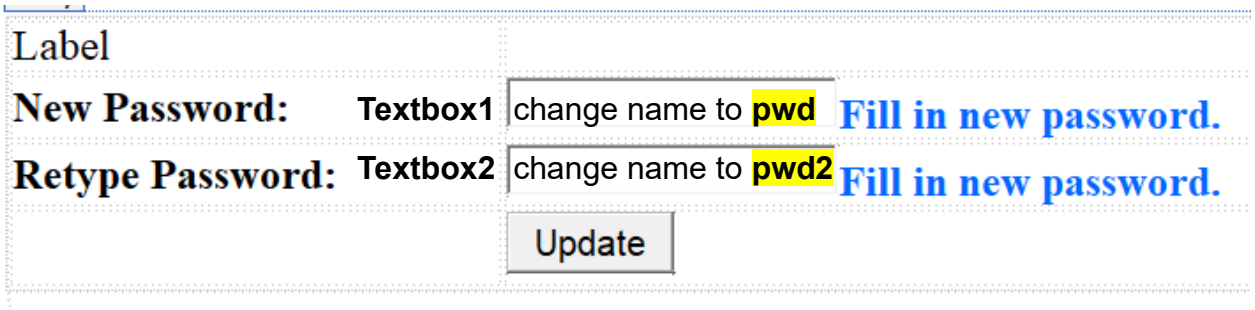
3. Type in the following code:-

```
protected void Button1_Click(object sender, EventArgs e)
{
    SqlConnection con = new SqlConnection(ConfigurationManager.ConnectionStrings["ConnectionString"].ConnectionString);
    con.Open();
    SqlCommand cmd = new SqlCommand("select Password from Users where Username = @Username", con);
    cmd.Parameters.AddWithValue("@Username", username.Text);
    object result = cmd.ExecuteScalar();
    if (result != null)
    {
        string storedHashedPassword = result.ToString();
        string userEnteredPassword = pwd.Text;

    // Verify the password
        bool isPasswordValid = BCrypt.Net.BCrypt.Verify(userEnteredPassword, storedHashedPassword);

        if (isPasswordValid)
        {
            // Password is correct, proceed with login
            Response.Write("Login successful!");
            SqlCommand cmdType = new SqlCommand("select Usertype from Users where Username = @Username", con);
            cmdType.Parameters.AddWithValue("@Username", username.Text);

            string type = cmdType.ExecuteScalar().ToString().Replace(" ", "");
            string uname = username.Text;

            Session["UserType"] = type;

            Session["Username"] = uname;
            if (type == "admin")
                Response.Redirect("adminPage.aspx");
            else if (type == "member")
                Response.Redirect("memberPage.aspx");

        }
    }
    else
    {
        this.errorMsg.Visible = true;
        this.errorMsg.ForeColor = System.Drawing.Color.Red;
        errorMsg.Text = "Username and Password mismatch!";
        return;
    }
    con.Close();
}
```

4. Save and test your function.

Change Password Page

5. Continue from the previous project
6. Create a new web form called ***changePassword.aspx***. See figure below.



7. Add in a label for greeting purpose (session variable for username).
8. Apply 2(two) required field validator to ensure these 2(two) textboxes are not empty.
9. Double click on ***Update button*** and type in the following code:-

```
using BCrypt.Net;
```

```
protected void Button1_Click(object sender, EventArgs e)
{

    if (pwd.Text == pwd2.Text)
    {
        //hashing
        var passwordManager = new PasswordManager();
        string hashedPassword = passwordManager.HashPassword(pwd.Text);

        SqlConnection con = new SqlConnection(ConfigurationManager.ConnectionStrings["ConnectionString"].ConnectionString);
        con.Open();
        //update first table
        string query = "update Users set Password = @Passowrd";
        SqlCommand cmd = new SqlCommand(query, con);
        cmd.Parameters.AddWithValue("@Passowrd", hashedPassword);
        cmd.ExecuteNonQuery();

        Response.Redirect("editProfile.aspx");
    }
}
```

```
public class PasswordManager
{
    // Method to hash the password
    1 reference
    public string HashPassword(string plainTextPassword)
    {
        string hashedPassword = BCrypt.Net.BCrypt.HashPassword(plainTextPassword);
        return hashedPassword;
    }

}
```

10. Save and test your function.